

Figure 4: The training and test loss of DMF with respect to the maximum iteration (T) on *Foursquare* and *Alipay* datasets.

Table 3: Performance comparison on *Alipay* dataset.

Metrics	P@5	R@5	P@10	R@10
$K=5$				
Dimension				
MF	0.0209	0.0896	0.0134	0.1194
BPR	0.0243	0.1027	0.0166	0.1408
GDMF	0.0209	0.0937	0.0144	0.1231
LDMF	0.0017	0.0042	0.0008	0.0042
DMF	0.0267	0.1128	0.0228	0.1824
$K=10$				
Dimension				
MF	0.0343	0.1493	0.0261	0.2239
BPR	0.0353	0.1512	0.0286	0.2439
GDMF	0.0304	0.1146	0.0277	0.2202
LDMF	0.0013	0.0065	0.0013	0.0131
DMF	0.0357	0.1612	0.0291	0.2537
$K=15$				
Dimension				
MF	0.0378	0.1538	0.0266	0.2308
BPR	0.0448	0.1898	0.0342	0.2872
GDMF	0.0355	0.1383	0.0267	0.2123
LDMF	0.0137	0.0684	0.0103	0.0983
DMF	0.0413	0.1942	0.0370	0.3035

choices of β and γ , DMF can make full use of one's own data and his neighbors' data, and thus, achieves the best performance.

Effect of maximum random walk distance (D). The maximum random walk distance determinates how many neighbors will be communicated after a user has interaction with a POI, as we described in complexity analysis section. Figure 6 shows its effect on DMF model on both *Foursquare* and *Alipay* datasets, where we set $K = 5$ and fix other parameters to their best values. From it, we see that with the increase of D , our model performance increases, and tends to be relative stable when D is bigger than 3. This shows that DMF achieves a good performance with only a small value of D , which indicates a low cost of communication complexity.

Effect of maximum iteration (T). As we analyzed above, the computing time complexity is linear with the training data size, and thus, the converge speed determines how long DMF should be trained. Figure 4 shows the effect of T on training loss and test loss on both *Foursquare* and *Alipay* datasets. As we can observe, DMF converges steadily with the increase of T , and it takes about 100 epochs to converge on *Foursquare* and about 200 epochs on *Alipay*.

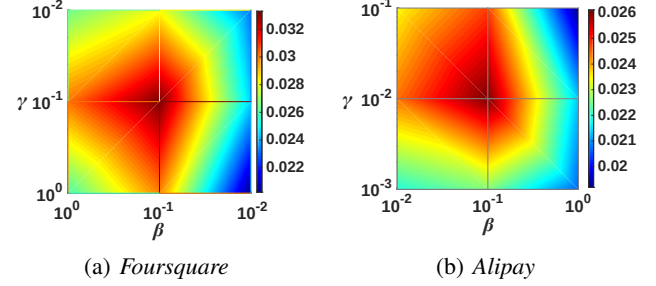


Figure 5: Effect of β and γ on DMF.

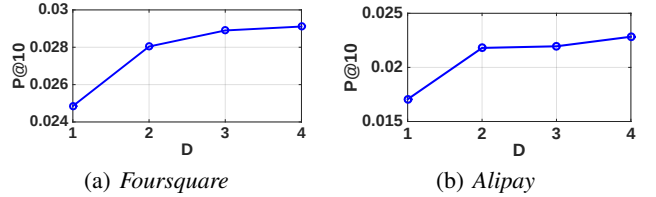


Figure 6: Effect of D on DMF.

Conclusion and Future Work

In this paper, we proposed a Decentralized MF (DMF) framework for POI recommendation. Specifically, we proposed a random walk based nearby collaborative decentralized training technique to train DMF model in each user's end. By doing this, the data of each user on items are still kept on one's own hand, and moreover, decentralized learning can be taken as distributed learning with multi-learner (user), and thus solves the efficiency problem. Experimental results on two real-world datasets demonstrate that, comparing with the classic and state-of-the-art latent factor models, DMF significantly improves the recommendation performance in terms of precision and recall.

We would like to take model compression as our future work. Currently, each user needs to store the real-valued item latent matrix. A binary type of latent matrix will significantly reduce the storage cost. How to balance the model storage and model accuracy will be our next stage of work.

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